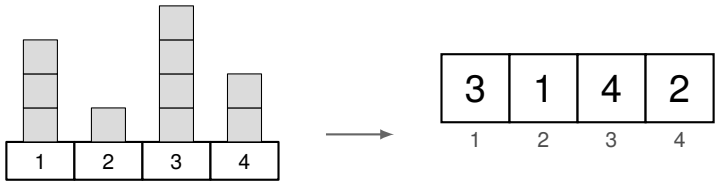
Build the towers on a strip of slots. Two players take turns. On your turn, take one whole tower from either end of the row. When the row is empty, the player with more cubes wins. The same number of cubes is a tie. A row is written as its tower heights from slot 1, so this row is 3, 1, 4, 2:



Playing as well as you can means collecting as many cubes as you can be sure of. Who wins a row, and by how many cubes, means the result when both players play as well as they can.

Problem 1: For each row, find who wins and by how many cubes. Write each player's total.

2, 1, 5, 3 first player _____ second player _____

1, 4, 6, 2 first player _____ second player _____

2, 4, 3, 1 first player _____ second player _____

6, 2, 1, 5 first player _____ second player _____

Problem 2: Who wins each row of three? Write each player's total. Find a rule that tells you who wins from the three heights, and explain why it works.

1, 5, 1 first player _____ second player _____

5, 1, 5 first player _____ second player _____

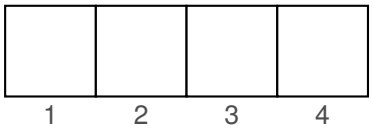
2, 6, 3 first player _____ second player _____

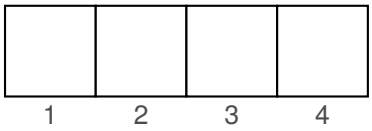
3, 4, 2 first player _____ second player _____

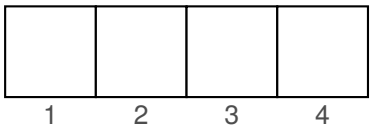
2, 5, 3 first player _____ second player _____

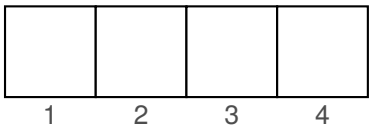
6, 1, 2 first player _____ second player _____

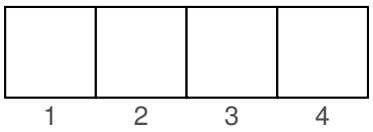
Problem 3: Is there a row of four towers, each 1 to 6 cubes high, on which the first player loses?

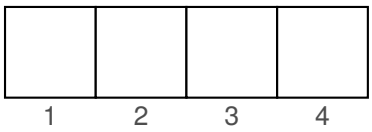


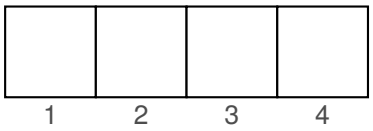


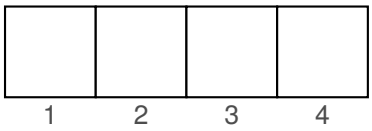




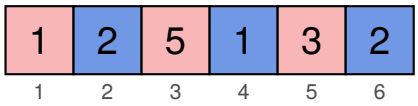


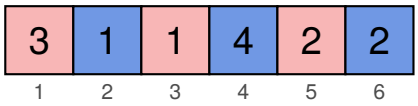


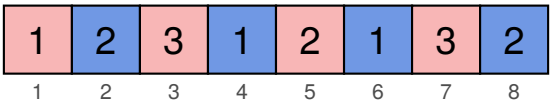




Problem 4: The grown-up goes first on the colored strips and never loses. Play these rows against the grown-up as a team, and work out the grown-up's plan.

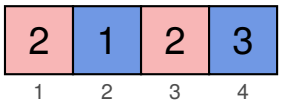




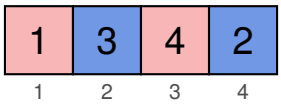


Problem 5: Explain why the grown-up's plan never loses on a row with an even number of towers, whatever the other player does.

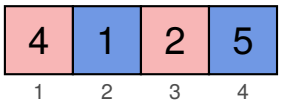
Problem 6: In each of these colored rows, the red slots and the blue slots hold the same number of cubes. Which rows can the first player win, and by how many cubes?



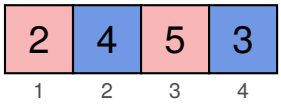
first _____ second _____



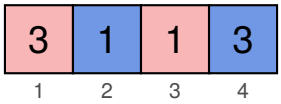
first _____ second _____



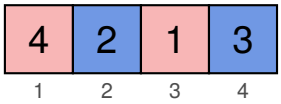
first _____ second _____



first _____ second _____

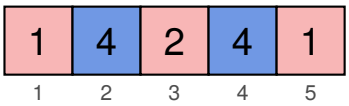


first _____ second _____

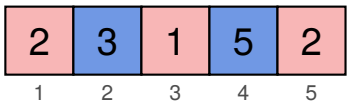


first _____ second _____

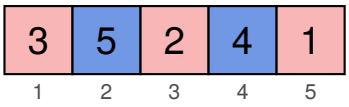
Problem 7: In each of these colored rows of five, the blue slots hold more cubes than the red slots. Who wins each row? Explain why.



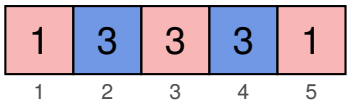
first _____ second _____



first _____ second _____

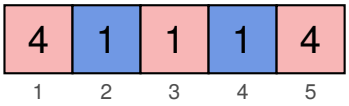


first _____ second _____

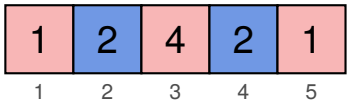


first _____ second _____

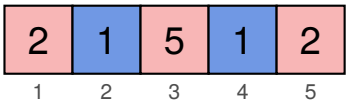
Problem 8: In each of these colored rows of five, the red slots hold more cubes than the blue slots. Who wins each row?



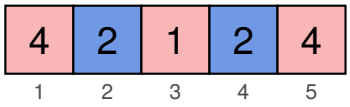
first _____ second _____



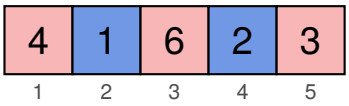
first _____ second _____



first _____ second _____



first _____ second _____



first _____ second _____

Problem 9: Can the first player win a row of five on which the red slots and the blue slots hold the same number of cubes? Explain.

Problem 10: For each row, ring the first player's best first move, and find who wins and by how many cubes.

3	1	4	1	5
1	2	3	4	5

first _____ second _____

2	1	2	5	3
1	2	3	4	5

first _____ second _____

5	2	6	1	3
1	2	3	4	5

first _____ second _____

Problem 11: In each box, write the most cubes the player whose turn it is can be sure of collecting from that row, in a game on 2, 3, 6, 2, 1, 4.

